

OUR FIRST QUERY 🔥

CREATE DATABASE db_name;

USE db_name;

DROP DATABASE db_name;

CREATE DATABASE IF NOT EXISTS db_name;

Create First Tables

```
CREATE TABLE table_name (  
    column1 datatype constraint,  
    column2 datatype constraint,  
    .....  
    columnN_name data_type constraint  
... );
```

Types of SQL Commands

SQL Commands are mainly categorized into five categories:

DDL (Data Definition Language) : **create, alter, rename, truncate & drop**

DQL (Data Query Language) : **select**

DML (Data Manipulation Language) : **insert, update & delete**

DCL (Data Control Language) : **grant & revoke permission to users**

TCL (Transaction Control Language) : **start transaction, commit, rollback (ACID)**

Data Definition Language (DDL)

"DDL sets up your database structure—tables, indexes, and schemas. It’s how you create, modify, or delete the blueprint of your database!"

Command	Description	Syntax
CREATE	Create database or its objects (table, index, function, views, store procedure, and triggers)	CREATE TABLE table_name (column1 data_type, column2 data_type, ...);
DROP	Delete objects from the database	DROP TABLE table_name;
ALTER	Alter the structure of the database	ALTER TABLE table_name ADD COLUMN column_name data_type;
TRUNCATE	Remove all records from a table, including all spaces allocated for the records are removed	TRUNCATE TABLE table_name;
COMMENT	Add comments to the data dictionary	COMMENT 'comment_text' ON TABLE table_name;
RENAME	Rename an object existing in the database	RENAME TABLE old_table_name TO new_table_name;

TEXT	Large text string (variable length)	description TEXT
LONGTEXT	Larger text data than TEXT	comments LONGTEXT
BLOB	Binary Large Object for storing images, files, etc.	image BLOB
DATE	Date in YYYY-MM-DD format	birthdate DATE
TIME	Time in HH:MM:SS format	start_time TIME
DATETIME	Date and time in YYYY-MM-DD HH:MM:SS format	created_at DATETIME
TIMESTAMP	Automatically tracks date and time of record creation/update	last_updated TIMESTAMP
YEAR	Stores a 4-digit year	join_year YEAR
BOOLEAN	Stores TRUE or FALSE values (often mapped to TINYINT(1))	is_active BOOLEAN
BINARY(n)	Fixed-length binary data (n bytes)	image_data BINARY(256)
VARBINARY(n)	Variable-length binary data (up to n bytes)	file_data VARBINARY(500)
ENUM	Predefined list of values, stores one of the possible options	gender ENUM('Male', 'Female', 'Other')
SET	Predefined list of values, stores multiple options (comma-separated)	tags SET('Science', 'Math', 'History')
JSON	Stores JSON (JavaScript Object Notation) formatted data	preferences JSON